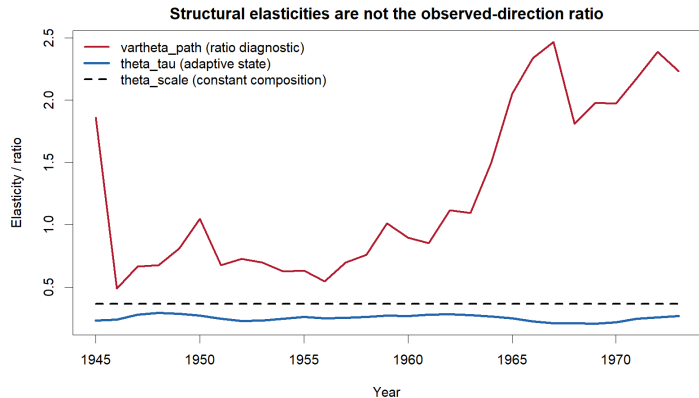


The instability appears after estimation, during reconstruction



The central distinction

The red line is not an equilibrium elasticity. It is capacity growth divided by aggregate capital growth in each observed year.

- The scale elasticity is constant within an estimated window.
- The technique elasticity moves with a wage-pressure state.
- The red ratio also moves with the annual mix and speed of capital accumulation.

Evidence source: `US_S90_path_dependent_theta_robustness.R` and `S90_reconstructed_path_roughness_comparison.csv`

Capacity can expand by enlarging plant or changing technique

Scale: more productive space

$$g_t^{NRC}$$

Structures measure the extension of plant, buildings, and the physical layout of production.

- Balanced accumulation expands machinery and structures together.
- Its capacity payoff is governed by $\theta^{scale} = \theta$.

Composition: more machinery per unit of plant

$$\Delta\tau_t = g_t^{ME} - g_t^{NRC}$$

Positive $\Delta\tau_t$ means that the installed technique becomes more machinery-intensive.

- Rapid mechanization changes the direction of accumulation.
- Its capacity payoff is governed by θ_t^τ .

The growth accounting identity

$$g_t^p = \theta^{scale} g_t^{NRC} + \theta_t^\tau (g_t^{ME} - g_t^{NRC})$$

Evidence source: US_S90_path_dependent_theta_robustness.R: corrected primitive-basis build_path

The three objects answer different economic questions

Object	Question answered	Economic status
$\theta^{scale} = \theta$	If structures and machinery expand proportionally, how much does capacity expand?	Equilibrium scale elasticity
$\theta_t^\tau = \psi + \lambda \tilde{d}_t$	At the current wage-pressure state, what is the capacity payoff from greater machinery intensity?	State-conditioned technique elasticity
$\vartheta_t^{path} = g_t^P / g_t^{cap}$	How much capacity growth occurred per unit of measured aggregate capital growth in this particular year?	Realized-direction ratio

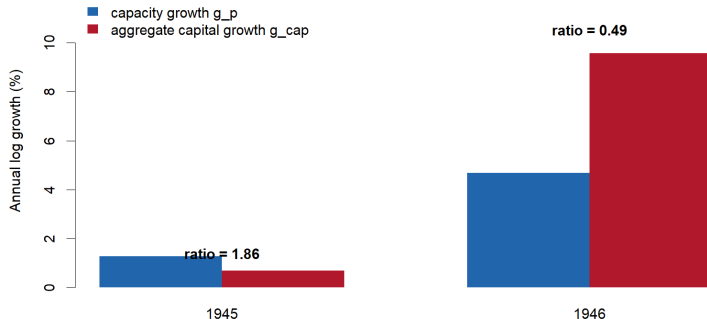
Interpretive rule

A volatile ϑ_t^{path} does not imply a volatile equilibrium frontier. It can move because the economy changes the composition of investment or because g_t^{cap} changes sharply.

Evidence source: `reconstruct_golden_age_paths.R` and `S90_original_vartheta_path_exact_contribution_ledger.csv`

Higher capacity growth can produce a lower capacity-payoff ratio

The ratio falls even though capacity growth accelerates



1945

$$g^p = 1.286\%$$

$$g^{cap} = 0.692\%$$

$$g^p / g^{cap} = \mathbf{1.86}$$

1946

$$g^p = 4.691\%$$

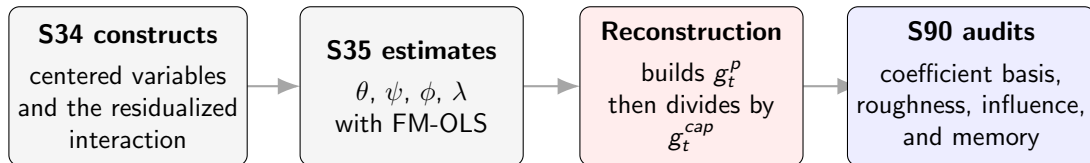
$$g^{cap} = 9.572\%$$

$$g^p / g^{cap} = \mathbf{0.49}$$

Capacity grows faster in the second year, but aggregate capital grows faster still. The ratio therefore falls.

Evidence source: D10 capital levels reproduced by US_S90_beamer_source_audit.R

The red path has a traceable four-script origin



Where the visual instability enters

S34 and S35 estimate a levels relationship. The annual red path is created later, when the reconstruction script converts those coefficients into capacity growth and divides by annual aggregate capital growth.

Evidence source: Source anchors and hashes: S90.beamer..script.provenance.ledger.csv

Orthogonalization relabels coefficients but leaves fitted capacity unchanged

Residual-centered interaction

$$r_t = (\tilde{\tau}_t \tilde{d}_t) - \delta_0 - \delta_k \tilde{k}_t^{NRC} - \delta_\tau \tilde{\tau}_t - \delta_d \tilde{d}_t.$$

The residual is orthogonal to the linear regressors, which improves numerical conditioning.

The economic derivative still requires the primitive basis

$$\psi^{primitive} = \psi^{orth} - \lambda \delta_\tau, \quad \theta^{primitive} = \theta^{orth} - \lambda \delta_k.$$

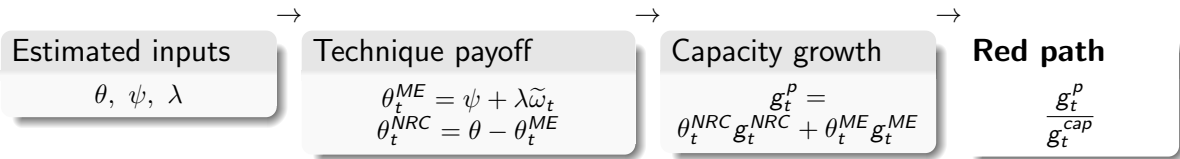
	orthogonal	primitive
θ	0.23784	0.34143
ψ	0.31207	0.27696
λ	-1.02937	-1.02937

Audit result

Fitted values agree to 4.8e-14. The back-transformation changes coefficient interpretation, not the fitted capacity schedule.

Evidence source: S34 full-panel residualization and S35 Golden-Age coefficients, independently reproduced in the Beamer claim ledger

The reconstruction script converts smooth effects into an annual flow ratio

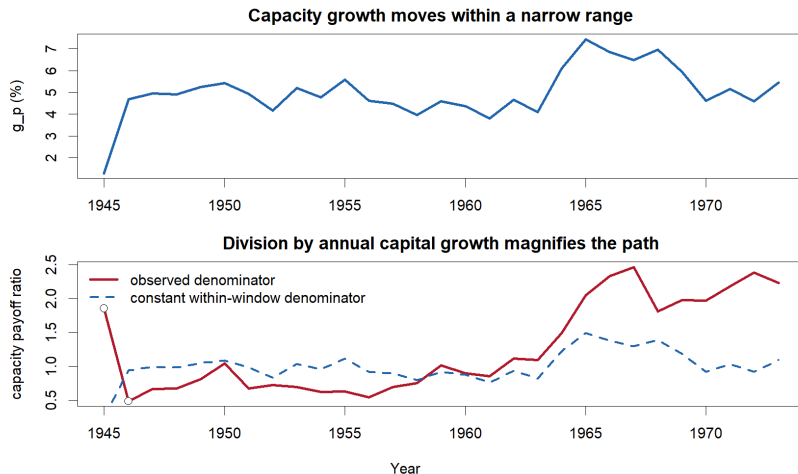


The label created the confusion

The script saves the final ratio as `theta_total_t`. That variable name suggests a structural elasticity even though the expression is an annual realized-capacity payoff.

Evidence source: `reconstruct_golden_age_paths.R`, source line 128; reconstruction series reproduced to numerical precision

Capacity growth is far smoother than the ratio formed from it



First-difference
roughness

g_t^p	0.0001008
red ratio	0.1236
fixed $\bar{g}^{cap}$	0.04053

Meaning

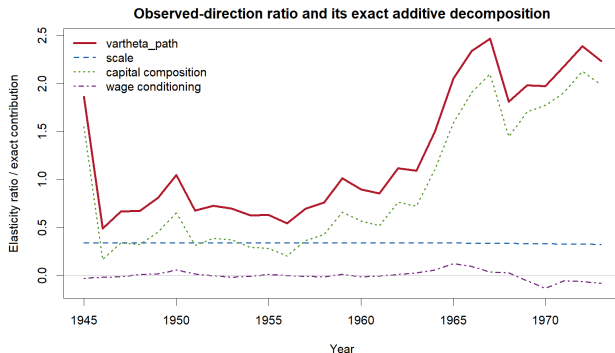
The denominator magnifies the path, but the fixed-denominator line still moves. Composition therefore remains part of the explanation.

Evidence source: S90_denominator_amplification_ledger.csv and independently reconstructed D10 growth rates

The exact decomposition isolates the economic channels

Exact identity

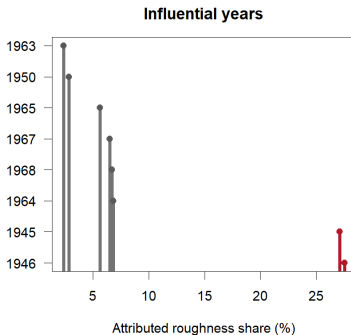
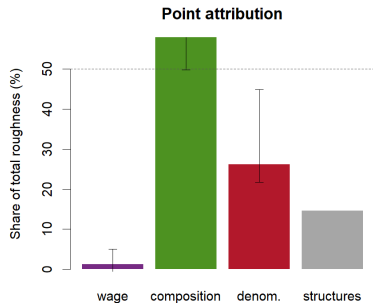
$$\vartheta_t^{path} = \frac{\theta g_t^{NRC}}{g_t^{cap}} + \frac{\psi(g_t^{ME} - g_t^{NRC})}{g_t^{cap}} + \frac{\lambda \tilde{d}_t(g_t^{ME} - g_t^{NRC})}{g_t^{cap}}.$$



The wage-conditioned contribution remains close to zero. The visible rise and fall follows the accumulation-composition term.

Evidence source: S90_original_vartheta_path_exact_contribution_ledger.csv; maximum identity error below numerical tolerance

Composition is point-dominant, but the evidence is not decisive



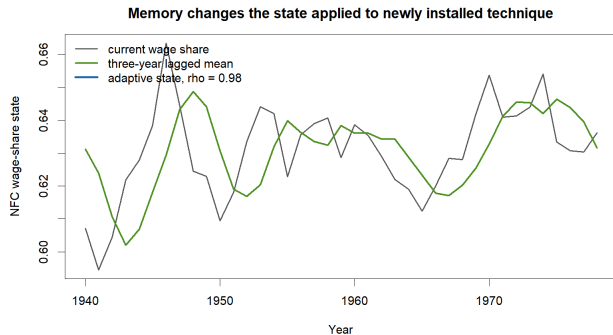
- Composition: **58.0%**
- Denominator: 26.2%
- Wage conditioning: 1.2%
- Structures reference: 14.6%

Two qualifications

The composition interval is 49.8-67.6%, crossing the majority threshold.
The 1945-1946 observations account for 54.7% of attributed roughness.

Evidence source: S90_shapley_roughness_attribution_ledger.csv and S90_capital_growth_and_influential_year_audit.csv

Path dependence attaches wage history to newly installed technique



Current state applies today's wage share to the entire technique stock.

Finite memory replaces today's share with a lagged moving average.

Embodiment weights each new change in machinery intensity by the wage state prevailing when it is installed.

Adaptive state estimates how quickly old wage information decays.

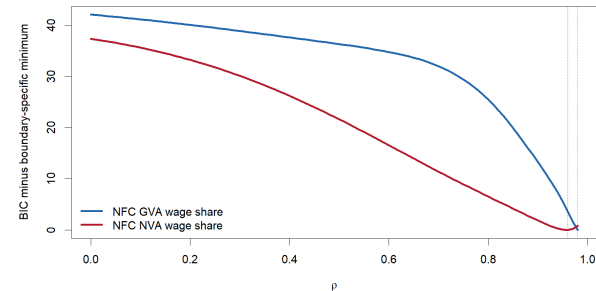
Accounting boundary

Output and capital remain NFC. Wage pressure is alternately measured for the whole NFC and all corporations; the latter tests a broader bargaining environment.

Evidence source: US_S90_path_dependent_theta_robustness.R: M0, M1, M2, and M3 state constructors

Persistence is strong; the technique effect is not stable

The NFC wage state adjusts slowly



wage state	$\hat{\rho}$	block interval	half-life
NFC GVA	0.98	0.92-0.98	≥ 34.3 years
NFC NVA	0.96	0.77-0.98	17.0 years

What survives

Embodied-memory M2 retains a negative Fordist conditioning effect under both memory lengths:

GVA: −28.00, −27.31
NVA: −12.76, −3.85.

Why the full claim fails

The GVA persistence estimate reaches the search boundary. M3’s conditioning coefficient also changes sign across GVA (60.54) and NVA (−12.43). Slow memory is evident; a stable equilibrium technique effect is not.

For v6: rename the red line, show g_t^P before division, and present memory as qualified evidence.

Evidence source: S90_M3_persistence_profile_ledger.csv and S90_RDOLS_coefficient_ledger.csv

A1. Every displayed result traces to a producing script and artifact

stage	producing script	audited artifact	MD5 prefix	status
S34 construction	run_s34r_b_cpr_realigned_design_gate.py	repaired augmented panel	afce5b84b7	PASS
S35 estimation	US_S35_estimator_refreeze_cpr.R	CPR estimation results	727e736cea	PASS
reconstruction	reconstruct_golden_age_paths.R	Golden-Age reconstructed paths	a263c1caef	PASS
S90 diagnostic	US_S90_path_dependent_theta_robustness.R	S90 coefficient, path, roughness, and persistence ledgers	2899921142	PASS

Audit contract

36 source-to-result claims were independently recomputed. Each row records the script expression, dynamically located source line, input and output artifacts, recomputed value, published value, tolerance, and pass/fail status.

Evidence source: S90_beamer_script_provenance_ledger.csv and S90_beamer_claim_reproduction_ledger.csv

A2. The coefficient-basis audit begins with the actual S34 and S35 code

S34 constructs the full-panel residualized interaction

```
# source: codes/S34R_B_cpr_realigned_design_gate/  
run_s34r_b_cpr_realigned_design_gate.py lines 368-370  
panel["inter_kKcap_omega_orth"] = residual_center(panel, "  
inter_kKcap_omega", ["k_Kcap_centered", "  
omega_NFC_centered"])  
panel["inter_kNRC_omega_orth"] = residual_center(panel, "  
inter_kNRC_omega", ["k_NRC_centered", "omega_NFC_centered"  
"])  
panel["inter_tau_omega_orth"] = residual_center(panel, "  
inter_tau_omega", ["k_NRC_centered", "tau_centered", "  
omega_NFC_centered"])
```

S35 sends that interaction to the FM-OLS estimator

```
# source: codes/US_S35_estimator_refreeze_cpr.R lines 80-85,  
126-128  
SPEC_B_NRC_orth = list(  
  rhs = c("k_NRC_centered", "tau_centered", "omega_NFC_centered"  
    , "inter_tau_omega_orth"),  
  coef_names = c("beta_scale", "beta_comp", "beta_dist", "  
    beta_inter"),  
  is_orth = TRUE  
)  
)  
  
if (estimator == "FM_OLS") {  
  cointReg::cointRegFM(x = x, y = y, deter = deter,  
    bandwidth = "nw")  
} else if (estimator == "IM_OLS") {
```

Primitive recovery

$$a^{raw} = a^{orth} - \lambda\delta_0, \quad \theta^{raw} = \theta^{orth} - \lambda\delta_k, \quad \psi^{raw} = \psi^{orth} - \lambda\delta_\tau, \quad \phi^{raw} = \phi^{orth} - \lambda\delta_d.$$

Independent residualization agrees with S34 to 6.8e-16; primitive and orthogonalized fitted values agree to 4.8e-14.

Evidence source: Source lines 370 and 127; coefficient values reproduced by US_S90_beamer_source_audit.R

A3. The two reconstruction boundaries are explicit script facts

Original reconstruction

```
# source: codes/reconstruct_golden_age_paths.R lines 48-50, 53-57, 62-62, 127-128
beta_k_B <- 0.23784
beta_tau_B <- 0.31207
beta_inter_B <- -1.02937
panel$theta_ME_t <- beta_tau_B + beta_inter_B * panel$omega_NFC_centered
panel$theta_NRC_t <- beta_k_B - panel$theta_ME_t

# Reconstruct potential output growth rate using the A03 composition identity
panel$g_Yp_B <- panel$theta_NRC_t * panel$g_NRC + panel$theta_ME_t * panel$g_ME
panel$yp_spec_B[idx_1973] <- panel$y_total[idx_1973]
# Calculate aggregate transformation elasticity theta_total = g_Yp_B / g_Kcap (
  growth-weighted average)
ga_data$theta_total_t <- ga_data$g_Yp_B / ga_data$g_Kcap
```

Original script: anchors productive capacity to total real GDP in 1973, while estimating elasticities from NFC variables.

S90 reconstruction

```
# source: codes/US_S90_path_dependent_theta_robustness.R lines 105-105,
  449-456, 506-513
panel$y <- log(panel$Y_REAL_NFC_GVA_BASELINE_D09)
primitive_beta <- c(
  intercept = orth_beta["intercept"] - orth_beta["lambda"] * delta["(Intercept)"]],
  theta = orth_beta["theta"] - orth_beta["lambda"] * delta["k_NRC_centered"],
  psi = orth_beta["psi"] - orth_beta["lambda"] * delta["tau_centered"],
  phi = orth_beta["phi"] - orth_beta["lambda"] * delta["omega_NFC_centered"],
  lambda = orth_beta["lambda"]
)
names(primitive_beta) <- names(orth_beta)
build_path <- function(df, beta, label) {
  theta_tau <- beta["psi"] + beta["lambda"] * df$d_c
  g_p <- beta["theta"] * df$g_nrc + theta_tau * (df$g_me - df$g_nrc)
  data.frame(year = df$year, reconstruction = label,
    theta_scale = unname(beta["theta"]), theta_tau = theta_tau,
    theta_me = theta_tau, theta_nrc = beta["theta"] - theta_tau,
    g_me = df$g_me, g_nrc = df$g_nrc, g_cap = df$g_cap,
    g_p = g_p, vartheta_path = g_p / df$g_cap,

```

S90 script: fixes observed output at real NFC GVA and anchors every reconstructed μ_{1973} to one.

Evidence source: Boundary claims SRC_RECON_OUTPUT_BOUNDARY and SRC_S90_OUTPUT_BOUNDARY in the claim ledger

A4. Reproduction succeeds before the expanded deck is compiled

Execution

Rscript codes/US_S90_beamer_source_audit.R

Required estimator package: cointReg.

Numerical checks

- 36 source-to-result claims: all PASS
- 8 S90 identities: all PASS
- 1000 moving-block draws per wage boundary
- coefficient tolerance: 5×10^{-5}
- algebraic identities: 10^{-10} or tighter

Generated interfaces

- script provenance ledger
- source-anchor ledger
- claim-reproduction ledger
- generated TeX number macros
- verbatim code excerpts

Preservation boundary

The audit writes only new S90 presentation assets. The v6 PDF, the six-slide addendum, S40, and Stage B/C remain unchanged.

Evidence source: S90_BEAMER_SOURCE_AUDIT.md and S90_validation_checks.csv